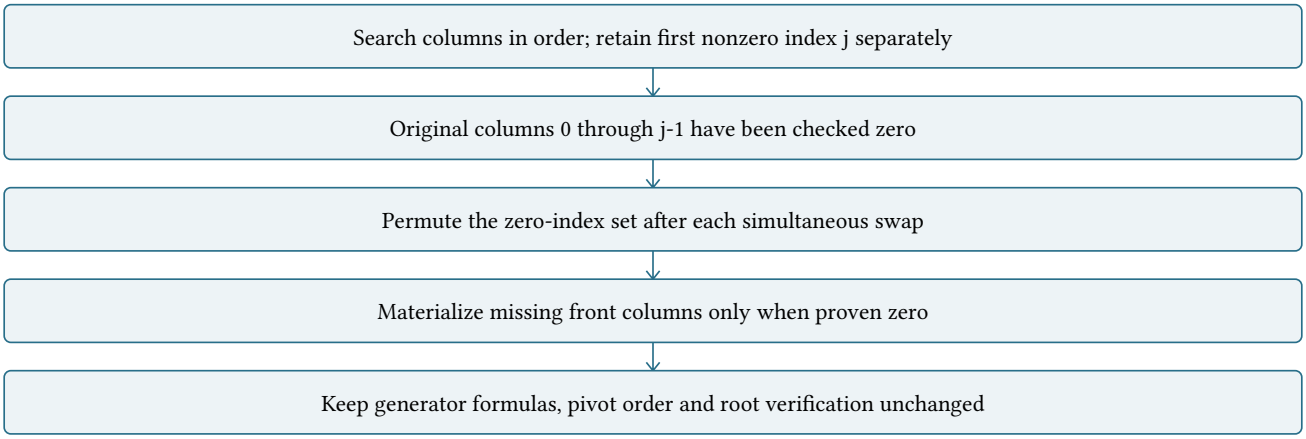
Proven-zero columns in paired transport

2026-11-02 · Derived schedule / static audit · Numerical trials: zero

Construction. The first nonzero column index j defines a checked zero prefix. Record it separately: a secondary diagonal pivot can be chosen at i greater than j. Preserve the original search and selected pivot.



Proof. If A e_t=0 and P swaps two coordinates, then (P A P)$e_{(p(t))}$ = P A e_t = 0. Thus the zero-index set follows the same permutation as the cached column keys. Missing front columns in this set can be materialized as zero tuples. The unchanged paired formulas still require the established weighted-symmetry invariant.

Derived case	Transport queries removed
Scalar pivot with j greater than 0	1
Pair pivot with j greater than 0	2
j equals 0	0
All-zero source	No pivot move

Width accounting. Each reused column removes one forward query at that swap's input width w, or 4w+8 instrumented native products under the preserved kernel schedule. For two pair swaps the widths are w and w+4: removed native products total 8w+32. Output width and all outside-query field arithmetic remain unchanged. Counts are derived; no elapsed-time comparison was run.

Costs and provenance. A set retains j indices and is permuted per swap. Every reuse materializes n zero entries. Native membership, canonical-cache, allocation and permutation work remains. New counters record these events; they do not measure total heap or runtime. Source identity, signature, exact record type, prefix range and one-use checks guard the internal handoff. They do not independently certify a forged prefix or mutated source.

Checks. Three complete declared-substitution AST comparisons preserve the selector, paired arithmetic and full metric-view driver. Five Python sources parse. Frozen references remain byte-identical. Dynamic correctness, corruption, counter, heap and performance validation remains unexecuted under the existing cost admission gate. No job or new build was launched.

Next construction. Use the same zero-coordinate information to derive coordinate deletion and primal/dual reinsertion, with a compatible displacement representation. That larger change is not implemented or included in counts.

Sources. selection_zero.py, paired_zero.py, driver_zero.py, integration.json, DERIVATION.md; unchanged finite-field pivot context: Dumas–Pernet2018, arXiv:1802.10453, introduction and Section 2.